

ORIGINAL RESEARCH ARTICLE

PUS7-Mediated Pseudouridylation of TGFBI Drives Vascular Remodeling in Pulmonary Hypertension

Junting Zhang¹, PhD*; Yiyang Li, PhD*; Muhua He, MS; Jianxin Tan², MS; Yuan Chen, MS; Zhongyuan Lin, MS; Hongbo Wang, MS; Fang Liu³, MS; Xiaolin Chen, BS; Zihui Jia, MS; Hanbin Chen, PhD; Zhen Chen, MS; Fanhao Kong, BS; Jiawen Fu, BS; Jin-Song Bian⁴, PhD; Xiao-Wei Nie⁵, PhD

BACKGROUND: Pulmonary hypertension (PH) is a life-threatening cardiovascular disorder characterized by irreversible pulmonary vascular remodeling and poor prognosis. RNA pseudouridylation, the most evolutionarily conserved RNA epigenetic modification, and its catalytic enzyme pseudouridine synthase 7 (PUS7) remained uncharacterized in PH, representing a major gap in the understanding of the epigenetic pathogenesis of the disease.

METHODS: We generated the first single-base resolution pseudouridine (Ψ) landscape in lung tissues of patients with PH using bisulfite-induced deletion sequencing. PUS7 expression was analyzed in hypoxic pulmonary artery endothelial cells, the lung tissues of patients with PH, and SU5416-hypoxia rodent model. The functional roles of PUS7 were investigated through genetic manipulation (PUS7-deficiency cells, adeno-associated virus serotype-mediated overexpression, endothelial cell-specific knockdown, and heterozygous knockout mice) and pharmacological inhibition with NSC107512.

RESULTS: Bisulfite-induced deletion sequencing revealed global Ψ dysregulation in the lung tissues of patients with PH. Among PUS family members, PUS7 was the most markedly upregulated in these tissues and in the hypoxic pulmonary artery endothelial cells. Both gene knockdown and pharmacological inhibition with NSC107512 ameliorated PH, whereas adeno-associated virus serotype-mediated PUS7 overexpression exacerbated disease progression. RNA immunoprecipitation sequencing and mutagenesis studies demonstrated that PUS7 bound to and catalyzed Ψ at position 688 of TGFBI (transforming growth factor β -induced protein) mRNA, thereby stabilizing TGFBI and activating phosphatidylinositol 3-kinase–protein kinase B signaling pathway. Furthermore, hypoxia-inducible factor 2 α bound directly to the PUS7 promoter, establishing a hypoxia-inducible factor 2 α /PUS7/TGFBI/phosphatidylinositol 3-kinase–protein kinase B positive feedback loop that drives PH pathogenesis.

CONCLUSIONS: PUS7-mediated pseudouridylation serves as a novel epigenetic driver of PH through the hypoxia-inducible factor 2 α /PUS7/TGFBI/phosphatidylinositol 3-kinase–protein kinase B axis, positioning PUS7 as a promising therapeutic target for this devastating disease.

Key Words: hypertension, pulmonary ■ PUS7 protein, human ■ TGFBI protein, human ■ vascular remodeling

Pulmonary hypertension (PH) is a life-threatening cardiovascular disorder characterized by progressive pulmonary vascular remodeling and elevated pulmonary vascular resistance.^{1,2} Under the classification

of World Symposium on Pulmonary Hypertension, PH secondary to chronic lung diseases or hypoxia is designated as group 3 PH. This subtype warrants urgent clinical attention because its prevalence continues to rise

Correspondence to: Xiao-Wei Nie, PhD, Department of Human Cell Biology and Genetics, School of Medicine, SUSTech Homeostatic Medicine Institute, School of Medicine, Southern University of Science and Technology, Shenzhen 518055, China. Email niexw@sustech.edu.cn

*J. Zhang and Y. Li contributed equally.

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Clinical Perspective

What Is New?

- By generating the first single-base resolution pseudouridine (Ψ) map, we revealed the critical role of global Ψ dysregulation in the pulmonary vascular remodeling of pulmonary hypertension (PH) patients and pseudouridine synthase 7 (PUS7) is identified as a PH epigenetic driver.
- A hypoxia-inducible factor 2 α (HIF-2 α)/PUS7/TGFBI/Phosphatidylinositol 3-Kinase (PI3K)-Protein Kinase B (AKT) positive feedback loop is identified in the PH pathogenesis: HIF-2 α activates PUS7 transcription, which stabilizes TGFBI mRNA via Ψ (position 688) to activate PI3K-AKT, driving progressive vascular remodeling.
- PUS7 is validated as an important therapeutic target with genetic and pharmacological manipulation.

What Are the Clinical Implications?

- PUS7 and TGFBI Ψ modification (position 688) may serve as a therapeutic target and a diagnostic biomarker for PH, respectively.
- PUS7 is enriched in PAECs, and the specific PUS7 inhibitor NSC107512 offers favorable therapeutic safety with minimal off-target effects.
- The HIF-2 α /PUS7/TGFBI/PI3K-AKT positive feedback loop provides a mechanistic explanation for the progressive nature of PH.

with increasing chronic lung disease burden, yet effective targeted therapies remain severely limited.³ For instance, chronic obstructive pulmonary disease, interstitial lung disease, and combined pulmonary fibrosis and emphysema collectively contribute to substantial increases in the prevalence of group 3 PH.^{4–6}

Pulmonary artery endothelial cell (PAEC) dysfunction is increasingly recognized to be a critical event in disease onset.^{7–9} Pathological stimuli such as hypoxia trigger PAEC dysfunction, promote proinflammatory and mitogenic signaling, and drive pulmonary artery smooth muscle cell (PASMC) proliferation.^{10,11} Despite recent advances, how the intricate molecular mechanisms and intercellular signaling networks govern hypoxia-induced PAEC dysfunction and their contribution to vascular remodeling remain poorly understood. It is therefore an urgent task to understand early mechanisms underlying PAEC dysfunction.

Pseudouridine (Ψ) is the most abundant posttranscriptional modification, distributed widely across diverse RNA species.¹² As opposed to the N1-C1 bond found in canonical uridine, Ψ is a C-C glycosidic isomer of uridine (U) with a distinct structure in which the uracil base is covalently attached to the ribose ring through a C5-C1 linkage.¹³ This structural divergence substantially modulates the physicochemical properties of Ψ , altering its

Nonstandard Abbreviations and Acronyms

AAV	adeno-associated virus serotype
AKT	protein kinase B
BID-seq	bisulfite-induced deletion sequencing
HIF-2α	Hypoxia-inducible factor 2 α
Ψ	pseudouridylation
PAAF	pulmonary artery adventitial fibroblast
PAEC	pulmonary artery endothelial cell
PASMC	pulmonary artery smooth muscle cell
PH	pulmonary hypertension
PI3K	phosphatidylinositol 3-kinase
PUS	pseudouridine synthase
RIP	RNA immunoprecipitation
RV	right ventricular
SI-PUS7	PUS7 siRNA
SuHx	SU5416/hypoxia
TGFBI	transforming growth factor- β induced protein
WT	wild type



base-stacking interactions and hydrogen-bonding patterns within the RNA helix, which enables Ψ to regulate the stability, structure, and function of these RNAs.^{13,14} The biological significance of Ψ remains largely unexplored. Elevated Ψ levels have been observed in biological samples from patients with cancer.^{15,16} Given that PH shares proliferative traits with cancer, Ψ is likely to influence PH pathogenesis. However, its contribution to hypoxia-induced vascular remodeling in PH has never been investigated.

Dysregulation of pseudouridylation is often driven by aberrant expression or activity of pseudouridine synthases (PUSs). Thirteen human PUSs, classified into 5 evolutionary families, catalyze Ψ formation.^{17,18} Among these PUSs, PUS7 was a conserved member of the PUS family, and our experimental data demonstrated that PUS7 expression is significantly upregulated in PH. PUS7 orchestrates codon-biased translation by catalyzing pseudouridylation of diverse RNA substrates, including mRNAs and transfer RNA fragments, thereby modulating translation efficiency.^{19,20} Functionally, its overexpression correlates with poor prognosis in bladder cancer, putatively by regulating immune cell infiltration within the tumor microenvironment.²¹ For example, PUS7 induces pseudouridylation of malignant T-cell–amplified sequence 1 mRNA to enhance its translation and drive oncogene activation.²² Additional studies demonstrate that PUS7 silencing potently suppresses colorectal cancer cell metastasis, whereas its overexpression enhances this process.²³ PUS7-mediated Ψ modification promotes cancer stem cell survival, and pharmacological inhibition of its enzymatic activity disrupts malignant transformation

in preclinical models.²⁴ Despite these advances, critical knowledge gaps persist concerning PUS7 in PH: Does it contribute to pseudouridylation imbalance in PH? The specific molecular mechanisms underpinning its function and the downstream signaling nodes remain to be elucidated.

In this study, we demonstrated that PUS7 was upregulated in patients with PH and in hypoxic PAECs. It catalyzed site-specific Ψ modification of TGFBI (transforming growth factor β -induced protein) and stabilized its mRNA expression, further activating a hypoxia-inducible factor 2 α (HIF-2 α)/PUS7/TGFBI/phosphatidylinositol 3-kinase (PI3K)–protein kinase B (AKT) positive feedback loop that exacerbated vascular remodeling. We further validated the critical role of PUS7 in the development of PH by pharmacological manipulation. Together, our findings provide the first evidence that PUS7-mediated pseudouridylation is a critical epigenetic driver of PH, offering a novel therapeutic target for this devastating disease.

METHODS

Data Availability Statement

The data and analytical methods supporting the findings of this study are available from the corresponding author on reasonable request. Detailed methods are provided in the [Supplemental Material](#).

Study Approval

All animal experimental protocols were reviewed and approved by the Committee of Southern University of Science and Technology. This study was conducted in strict accordance with the *Guide for the Care and Use of Laboratory Animals* published by the US National Institutes of Health (publication No.85-23, revised 1996). All human samples were collected with the informed consent of the patients, and the study procedures were approved by the Ethics Committee for the Use of Human Subjects of the Southern University of Science and Technology (China), which was in accordance with the Code of Ethics of the Helsinki Declaration of the World Medical Association for experiments involving humans.

Quantification and Statistical Analysis

Representative figures/images reflected the average level of each experiment. Normality of the data (using the Shapiro-Wilk test) and the equality of group variance (using the Brown-Forsythe test) were performed on all data with GraphPad Prism 9.0 software (GraphPad Software Inc, La Jolla, CA). All data collection and processing in this study are expressed as mean $\pm$ SD. An unpaired 2-tailed *t* test was used for data involving 2 groups with a normal distribution. When the data were nonnormally distributed, the nonparametric Mann-Whitney *U* test was applied. In the analysis of data from >2 groups, 1-way ANOVA followed by the Tukey honestly significant difference multiple-comparison test was used when the assumptions satisfied a normal distribution unless otherwise specified in figure

legends. When the assumptions did not satisfy a normal distribution, we used the nonparametric Kruskal-Wallis test with false discovery rate to correct for multiple comparisons. If there were 2 variables, 2-way ANOVA with the Tukey honestly significant difference test for multiple comparisons was used. In this article, values of “*n*” denote the number of independent biological replicates (the real number of biological samples or the real number of donors/animals). Technical replicates from the same donor were averaged before statistical analysis. A result of $P<0.05$ was considered statistically significant.

RESULT

Ψ Modification is Significantly Dysregulated in PH

To characterize single-base resolution changes in Ψ modification landscapes in PH, we performed bisulfite-induced deletion sequencing (BID-seq) in the lung tissues from 5 patients with PH (group 3) and 5 control donors. Transcriptomic analysis showed a robust landscape of Ψ -modified mRNAs in patients with PH (Figure 1A). Motif distribution profiles revealed that flanking sequences of mRNA with Ψ modifications were enriched with GCTA and GCAT motifs (Figure 1B). High modification frequency was found with UG Ψ GU, AG Ψ GG, UG Ψ GG, and GG Ψ GG motifs (Figure 1C). Ψ modification probability plots demonstrated many sites in the control concentrated at a Ψ ratio of 0. In contrast, PH exhibited a higher distribution frequency within the Ψ ratio of 0 to 0.25 compared with control, indicating higher Ψ levels in PH (Figure 1D). In addition, 60.8% of modifications were found to be localized in the coding regions (CDS; Figure 1E). Enrichment analysis based on Kyoto Encyclopedia of Genes and Genomes and Gene Ontology databases revealed that these genes were enriched in extracellular matrix–receptor interactions, cell adhesion molecules, PI3K-AKT signaling, and angiogenesis (Figure 1F and 1G). Gene set enrichment analysis further showed significant enrichment in response to transforming growth factor- β and extracellular matrix organization pathways (Figure 1H), suggesting that dysregulated Ψ modification may modulate PH pathogenesis. We therefore further assessed the expression of functionally characterized PUSs. Quantitative real-time polymerase chain reaction confirmed that PUS7 was markedly upregulated in the lung tissues of patients with group 3 PH, with greater upregulation than other PUSs (Figure 1I–1M).

PUS7 Upregulation in Lung Tissues and Pulmonary Arteries of PH Models and Patients

Western blotting analysis showed PUS7 upregulation in the lung tissues of hypoxia combined with SU5416 (SuHx) PH mouse/rat models, as well as in patients with group 1 and group 3 PH (Figure 2A–2C and [Figure S1A](#)).

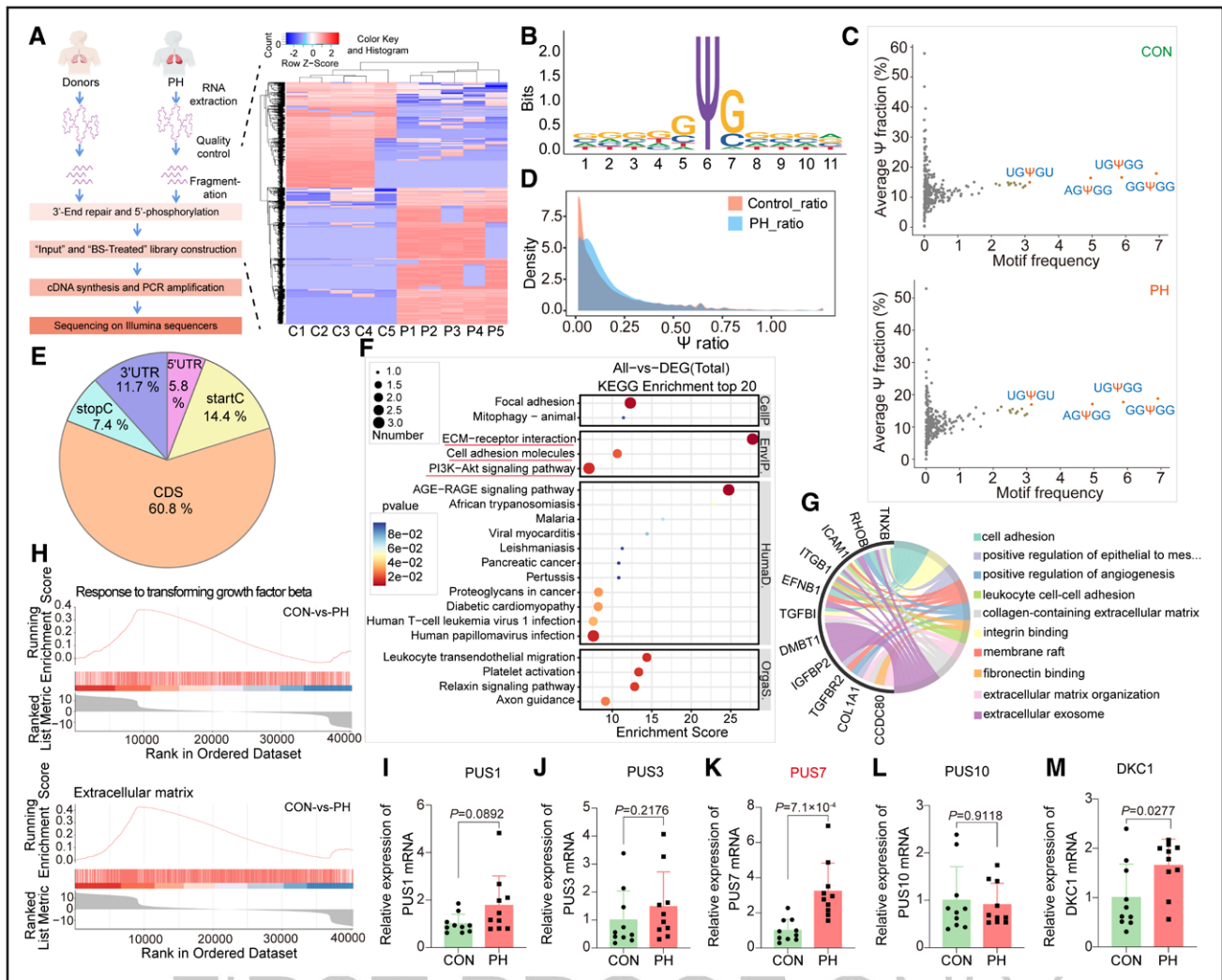


Figure 1. Ψ modification in the lung tissues of PH.

A, Bisulfite-induced deletion sequencing (BID-seq) showing pseudouridine (Ψ) modification levels in mRNAs in the lung tissues of patients with group 3 pulmonary hypertension (PH; $n=5$ patients). **B**, Motif distribution profiles of Ψ -modified mRNAs in the PH groups. **C**, Scatterplot illustrates the frequency of Ψ -modified motifs in donors and patients with PH. **D**, Ψ modification probability plots. The x axis represents Ψ ratio; the y axis represents the probability density for a specific ratio. Higher probability density indicates a greater likelihood of that ratio in the respective group. **E**, Frequency of Ψ modifications across distinct regions of mRNAs. **F** through **H**, Kyoto Encyclopedia of Genes and Genomes (KEGG), Gene Ontology (GO), and Gene Set Enrichment Analysis (GSEA; **H**) analyses were employed to elucidate the biological functions and signaling pathways associated with Ψ -modified mRNAs. **I** through **M**, Quantitative real-time polymerase chain reaction analysis showing the expression of pseudouridine synthases in the lung tissues from patients with group 3 PH ($n=10$ patients). CON (C), Control; PH (P), Pulmonary hypertension; Statistical analysis was performed with a 2-tailed Mann-Whitney test (**I**, **J** and **L**) or unpaired 2-tailed Student t-test (**K** and **M**); All values are presented as mean $\pm$ SD. AGE indicates advanced glycation end-products; DEG, differentially expressed gene; ECM, extracellular matrix; PUS, pseudouridine synthase; and RAGE, receptor for AGE.

Immunohistochemical and immunofluorescence staining confirmed marked upregulation of PUS7 with predominant localization to the pulmonary vascular intima (Figure 2D–2F and Figure S1B). Single-cell RNA sequencing of lung tissues from patients with group 3 PH demonstrated that PUS7 is specifically and significantly upregulated in endothelial cells from patients with group 3 PH (Figure 2G and 2H), mirroring the expression pattern observed in public database (GSE210248, GSE169477, and GSE169471) analyses of patients with pulmonary arterial hypertension (Figure S1C and S1D),

confirming consistent upregulation of PUS7 in PAECs across distinct PH subgroups. Western blotting analysis validated a pronounced upregulation of hypoxia-induced PUS7 in primary PAECs and pulmonary artery adventitial fibroblasts (PAAFs), but not in PASMCs (Figure 2I–Figure 2K). In cells from patients with group 3 PH, PUS7 expression was higher in PAECs, with relatively lower levels detected in PAAFs and PASMCs (Figure S1E). Immunostaining analysis showed the increased expression of PUS7 in both nuclear and cytoplasmic fractions in PAECs in a hypoxic situation (Figure 2L). It

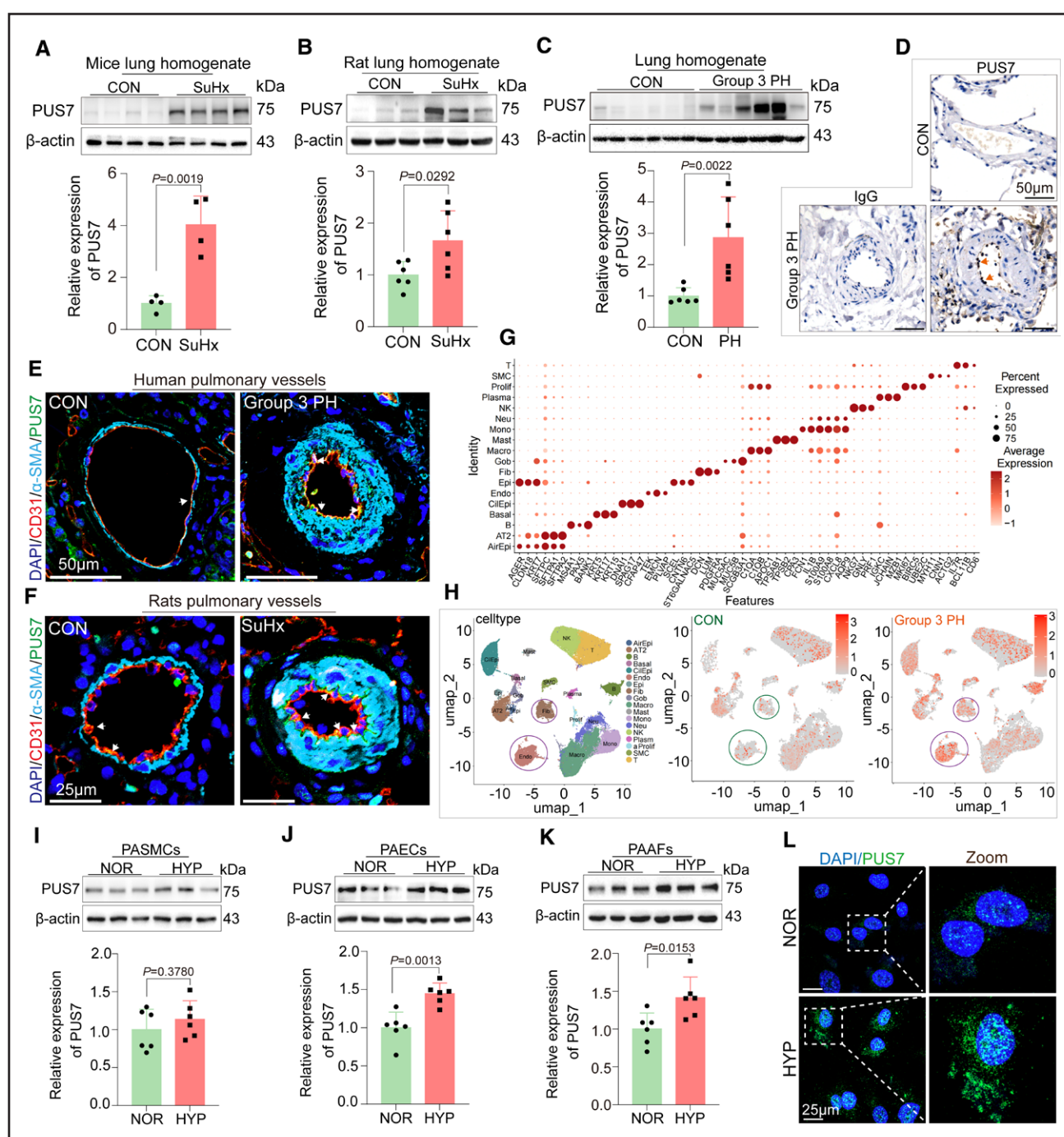


Figure 2. PUS7 was upregulated in lung tissues and pulmonary arteries in PH animal models and patients.

A through **C**, Representative Western blots showing the expression of pseudouridine synthase 7 (PUS 7) in the lung tissues of SU5416/hypoxia (SuHx)–induced rodent pulmonary hypertension (PH) models (**A**, $n=4$ male mice; **B**, $n=6$ male rats) and patients with PH (**C**, $n=6$ patients). **D**, Immunohistochemistry showing the expression of PUS7 in the lung tissues of healthy donors and patients with PH. **E** and **F**, Representative confocal images of immunohistochemical staining in lung sections from patients with PH (**E**) or SuHx-treated rats (**F**). Nuclei were stained with DAPI (blue). **G**, Dot plot of marker genes expressed in cell clusters, identified with the Find All Markers function. **H**, UMAP visualization depicts cell clustering and the distribution of PUS7 gene expression across group 3 PH and control (CON) groups. **I** through **K**, Representative Western blots and quantification of PUS7 and β -actin in the hypoxic pulmonary artery smooth muscle cells (PAMSCs; **I**), pulmonary artery endothelial cells (PAECs; **J**), and PAAFs (pulmonary artery adventitial fibroblasts; **K**; $n=6$ biological replicates). **L**, Representative confocal images of immunohistochemical staining for PUS7 (green) in hypoxic PAECs. Nuclei were stained with DAPI (blue). Statistical analysis was performed with a 2-tailed Mann-Whitney test (**C**) or unpaired 2-tailed Student t-test (**A**, **B**, **I**, **J**, and **K**). All values are presented as mean $\pm$ SD. HYP indicates hypoxia; and NOR, normoxia.

is important to note that ELISA demonstrated that serum PUS7 levels were significantly elevated in patients with group 3 PH compared with controls, suggesting that circulating PUS7 may serve as a minimally invasive diagnostic indicator for PH (Figure S1F). These results suggest that PUS7 may potentially be involved in the regulation of cellular functions under hypoxic stress.

PUS7 Modulates the Development of PH in Animal Models

Because of homozygous lethality of systemic PUS7 knockout (<https://mousephenotype.org/data/genes/MGI:1925947>), PUS7 heterozygous (PUS7^{+/-}) and littermate control (PUS7^{+/+}; wild type [WT]) mice were used to assess PUS7 biological functions in SuHx-induced PH (Figure 3A). Western blotting analysis confirmed PUS7 loss efficiency (Figure 3B). PUS7 deficiency did not significantly affect mouse body weight in both normal and SuHx/bleomycin treatment conditions (Figure 3C and Figure S2A). Treatment with SuHx or bleomycin significantly impaired pulmonary artery velocity-time integral, tricuspid annular plane systolic excursion, pulmonary artery acceleration time/ejection time ratio, elevated right ventricular (RV) systolic pressure and RV weight to left ventricular plus septal weight in PUS7^{+/+} mice. However, these effects were ameliorated in PUS7^{+/-} mice (Figure 3D–3I and Figure S2B–S2G). Similarly, compared with their littermate controls, PUS7-deficiency mice also displayed thinner vascular media and RV wall, lower α -smooth muscle actin expression, and fewer muscularized pulmonary arteries on SuHx or bleomycin treatment (Figure 3J–3O and Figure S2H–S2K). In addition, in vivo PUS7 overexpression alone elicited mild pulmonary hypertensive pathology under normoxia, and these pathological alterations were substantially aggravated in PH model animals (Figure S3).

Targeted Silencing of PUS7 in PAECs Alleviates Hypoxia- or Bleomycin-Induced PH

To define the role of endothelial PUS7 in PH progression, a Tie1 promoter-driven adeno-associated virus serotype (AAV) vector was used for AAV-ECKD (endothelial cell-specific PUS7 knockdown; Figure 4A). Echocardiography revealed that treatment with hypoxia or bleomycin significantly impaired pulmonary artery velocity-time integral, pulmonary artery acceleration time/ejection time ratio, and tricuspid annular plane systolic excursion in AAV-negative control mice. In contrast, these pathological effects were markedly attenuated in AAV-ECKD mice (Figure 4B–4D). Endothelial cell-specific PUS7 silencing also attenuated hypoxia- or bleomycin-induced elevations in RV systolic pressure and RV/left ventricular plus septal

weight ratio (Figure 4E–4G). Consistently, morphological analysis revealed that AAV-ECKD markedly attenuated hypoxia- or bleomycin-induced pulmonary artery muscularization and vascular media thickening compared with AAV-negative control (Figure 4H and 4I). Collectively, these findings demonstrate that endothelium-specific silencing of PUS7 in vivo markedly attenuates hypoxia- or bleomycin-induced PH pathogenesis in mice.

PUS7 Modulates Pathological Dysfunction of PAECs and PAAFs

To evaluate PUS7 function in PAECs, we designed PUS7 siRNA (SI-PUS7) and confirmed with quantitative real-time polymerase chain reaction and Western blotting analyses (Figure 5A and Figure S4A). EdU, Western blotting, and CCK8 (Cell Counting Kit-8) assays showed that SI-PUS7 significantly alleviated hypoxia-induced excessive proliferation (Figure 5B–5G and Figure S4B). Flow cytometry showed that SI-PUS7 prevented the G1-to-S phase transition (Figure 5H) and increased apoptosis in hypoxic PAECs (Figure 5I). Transwell (Figure 5J and 5K) and tube formation (Figure 5L and 5M) assays demonstrated that SI-PUS7 mitigated hypoxia-induced excessive migration and enhanced angiogenic capacity in PAECs. PUS7 also notably altered cytoskeletal architecture. In hypoxic human PAECs, stress fibers appeared disorganized, and paxillin displayed a less distinct distribution pattern. However, SI-PUS7 reversed the hypoxia-induced abnormalities, restoring well-defined stress fibers (Figure 5N) and proper localization of paxillin at focal adhesions (Figure 5O). Of note, similar effects were observed in PAAFs, in which PUS7 silencing abrogated hypoxia-induced PAAFs excessive proliferation (Figure S5). In addition, sustained PUS7 overexpression triggers profound dysfunction in PAECs under normoxia (Figure S6). These results suggest that PUS7 in PAECs plays a crucial role in the development of hypoxia-induced PH.

TGFBI Is a Key Downstream Gene of PUS7 in PH

RNA high-throughput transcriptome sequencing was performed to identify affected genes in PUS7-knockdown PAECs (Figure 6A and 6B). Because Ψ modification enhances RNA stability, we identified overlapping genes between downregulated genes in RNA high-throughput transcriptome sequencing (PUS7 knockdown) and genes harboring Ψ modification sites exclusively in patients with PH in BID-seq. Intersection of the 2 omics datasets identified 24 overlapping genes (Figure 6C). Using gene set enrichment analysis and enrichment analysis based on Kyoto Encyclopedia

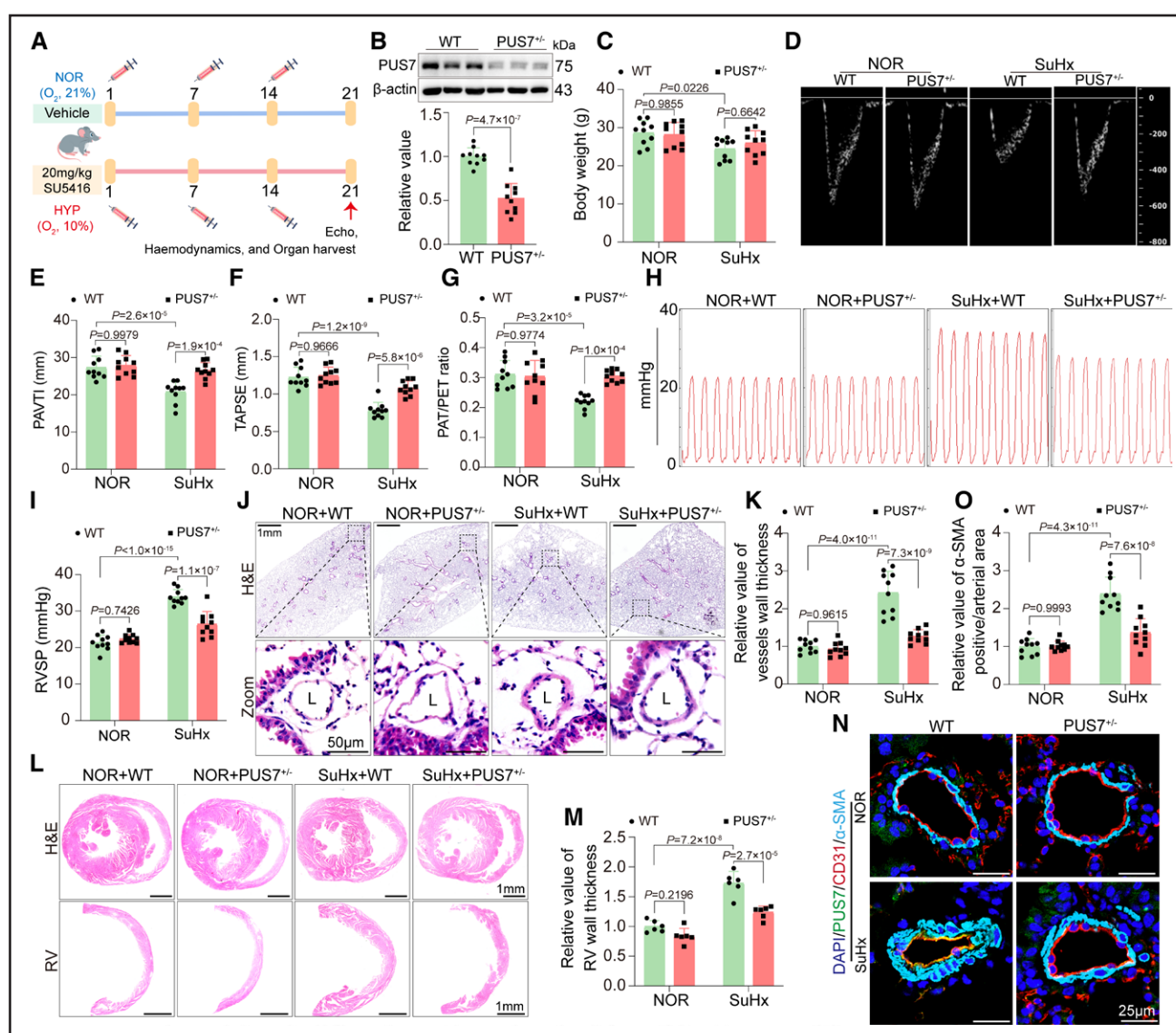


Figure 3. PUS7 loss ameliorated SuHx-induced PH in mice.

A, Schematic protocol for the animal treatments in the chronic hypoxia combined with SU5416 (SuHx)–induced pulmonary hypertension (PH) model. Pseudouridine synthase 7 (PUS7)^{+/+} (wild-type [WT]) and PUS7^{+/-} mice were placed in hypoxia for 21 days with 3 injections of SU5416 (20 mg/kg) on days 1, 7, and 14. **B**, Representative Western blots and quantification of PUS7 and β-actin in the lungs of WT and PUS7^{+/-} mice (n=10 mice, 6 males and 4 females). **C**, Group analysis showing that knockdown of PUS7 had no significant effect on mouse body weight (n=10 mice, 6 males and 4 females). **D** through **G**, Representative echocardiogram images (**D**) and group data showing PUS7^{+/-} attenuated the decreased pulmonary artery velocity-time integral (PAVTI; **E**), tricuspid annulus plain systolic excursion (TAPSE; **F**), and pulmonary artery acceleration time/ejection time ratio (PAT/PET; **G**) in mice treated with SuHx (n=10 mice, 6 males and 4 females). **H** and **I**, Representative hemodynamic tracings and group analysis showing the effects of PUS7^{+/-} on right ventricular (RV) systolic pressure (RVSP) of the SuHx mouse model (n=10 mice, 6 males and 4 females). **J** through **O**, Hematoxylin-eosin (H&E) and immunofluorescence staining showing that knockdown of PUS7 significantly ameliorated the SuHx-induced thickened medial wall (**J** and **K**) and right ventricular wall (**L** and **M**), and vascular muscularization (**N** and **O**) in the SuHx mouse model (n=10 mice, 6 males and 4 females). Statistical analysis was performed with 2-way ANOVA followed by the Tukey honestly significant difference test for multiple comparisons (**C**, **E** through **G**, **I**, **K**, **M**, and **O**) or unpaired 2-tailed Student *t* test (**B**). All values are presented as mean±SD. L indicates lumen; and NOR, normoxia.

of Genes and Genomes and Gene Ontology databases (Figure S7A–S7C), we selected 4 genes (*TGFBI*, *IL6ST*, *BCL2L12*, and *TACC1*) for further validation. Quantitative real-time polymerase chain reaction showed that PUS7 knockdown significantly reduced the expression of *TGFBI*, *BCL2L12*, and *TACC1*, with the most prominent reduction in *TGFBI* (Figure 6D). We then focused

on *TGFBI* as a downstream target of PUS7. Immunohistochemistry, immunofluorescence, and Western blots further confirmed upregulation of *TGFBI* in the lung tissues of patients with PH (Figure 6E–6G) and SuHx mice (Figure 6H). Furthermore, Western blots verified that PUS7 knockdown modulated the expression of *TGFBI* in PAECs (Figure 6I). Fluorescence in situ hybridization

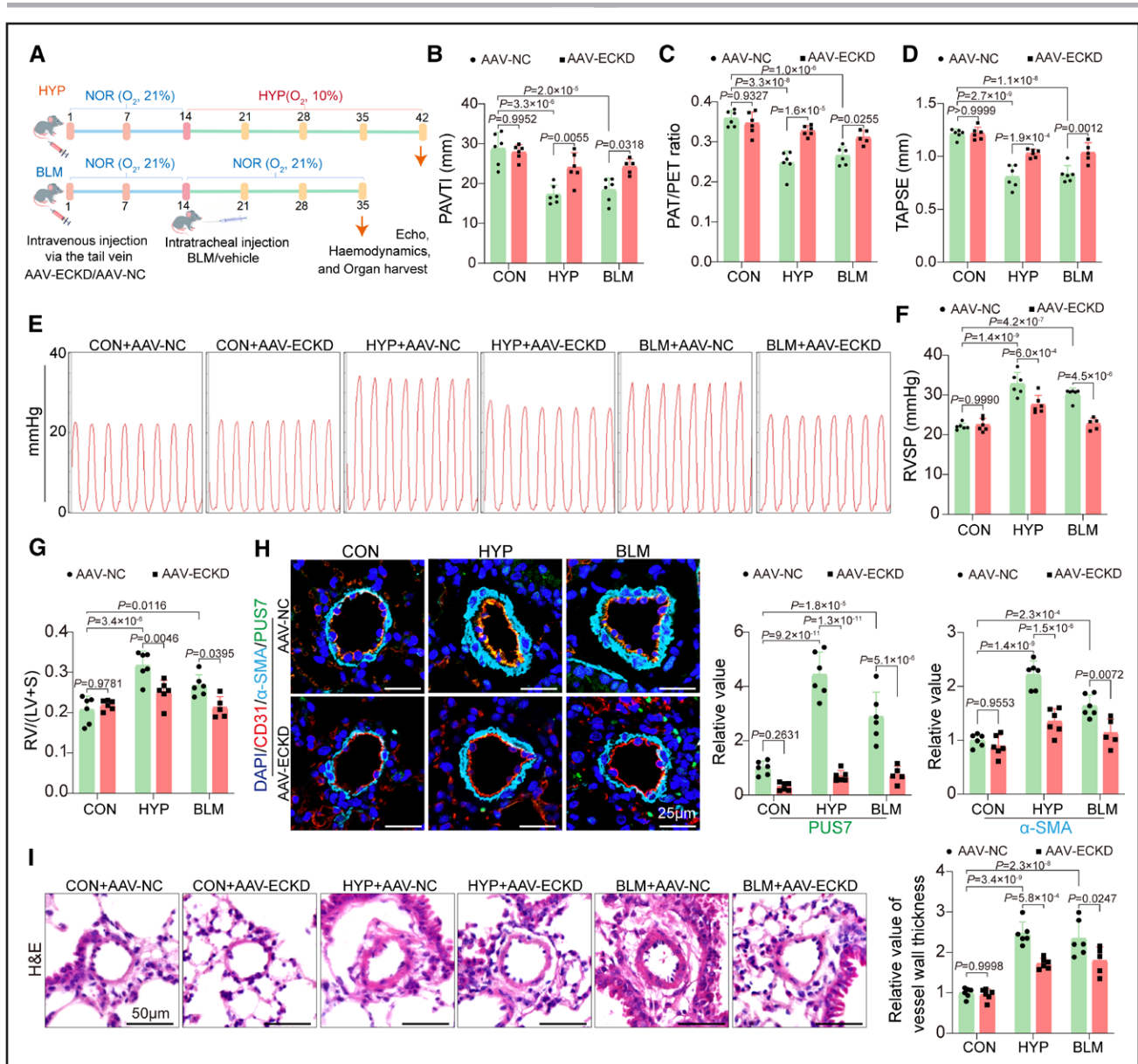


Figure 4. Endothelial cell-specific silencing of PUS7 attenuates hypoxia- or bleomycin-induced PH in mice.

A, Schematic showing the protocol. Mice received a single-bolus tail vein injection with Tie1 promoter-driven adeno-associated virus vector LungX for endothelial-specific knockdown pseudouridine synthase 7 (PUS 7) (AAV-ECKD) or AAV-negative control (NC), were placed in normoxic (NOR) conditions for 14 days, and then were subjected to hypoxia (HYP) for 28 days or administered bleomycin (BLM) intratracheally and maintained under normoxia for 21 days. **B** through **D**, Group analysis showing the decreased pulmonary artery velocity-time integral (PAVTI; **B**), pulmonary artery acceleration time/ejection time ratio (PAT/PET; **C**), and tricuspid annulus plain systolic excursion (TAPSE; **D**) in hypoxia- or BLM-induced pulmonary hypertension (PH) mouse model, and these parameters were improved in AAV-ECKD mice (n=5–6 mice, 3 males and 2–3 females). **E** through **G**, Representative hemodynamic tracings and group analysis showing the effects of endothelium-specific PUS7 silencing on right ventricular systolic pressure (RVSP; **E**; **F**, Statistical analysis) and ratio of right ventricle to left ventricle plus septum weight [RV/(LV+S); **G**] in hypoxic or BLM-treated mouse models of PH (n=5–6 mice, 3 males and 2–3 females). **H** and **I**, Immunofluorescence staining and hematoxylin-eosin (H&E) showing that knockdown of PUS7 significantly ameliorated hypoxia- or BLM-induced increases in vascular muscularization (**H**) and medial wall thickness (diameter 25–75 μ m; **I**; n=5–6 mice, 3 males and 2–3 females). Statistical analysis was performed with 2-way ANOVA followed by the Tukey honestly significant difference test for multiple comparisons (**B** through **I**); All values are presented as mean $\pm$ SD.

assay showed obvious colocalization of PUS7 with TGFBI mRNA (Figure 6J). RNA immunoprecipitation (RIP) assay and RIP sequencing demonstrated that PUS7 bound directly to TGFBI mRNA (Figure 6K and

Figure S7D). Actinomycin D assay showed that PUS7 overexpression reduced TGFBI mRNA degradation (Figure 6L). Collectively, these findings suggest that PUS7 regulates TGFBI mRNA expression directly.

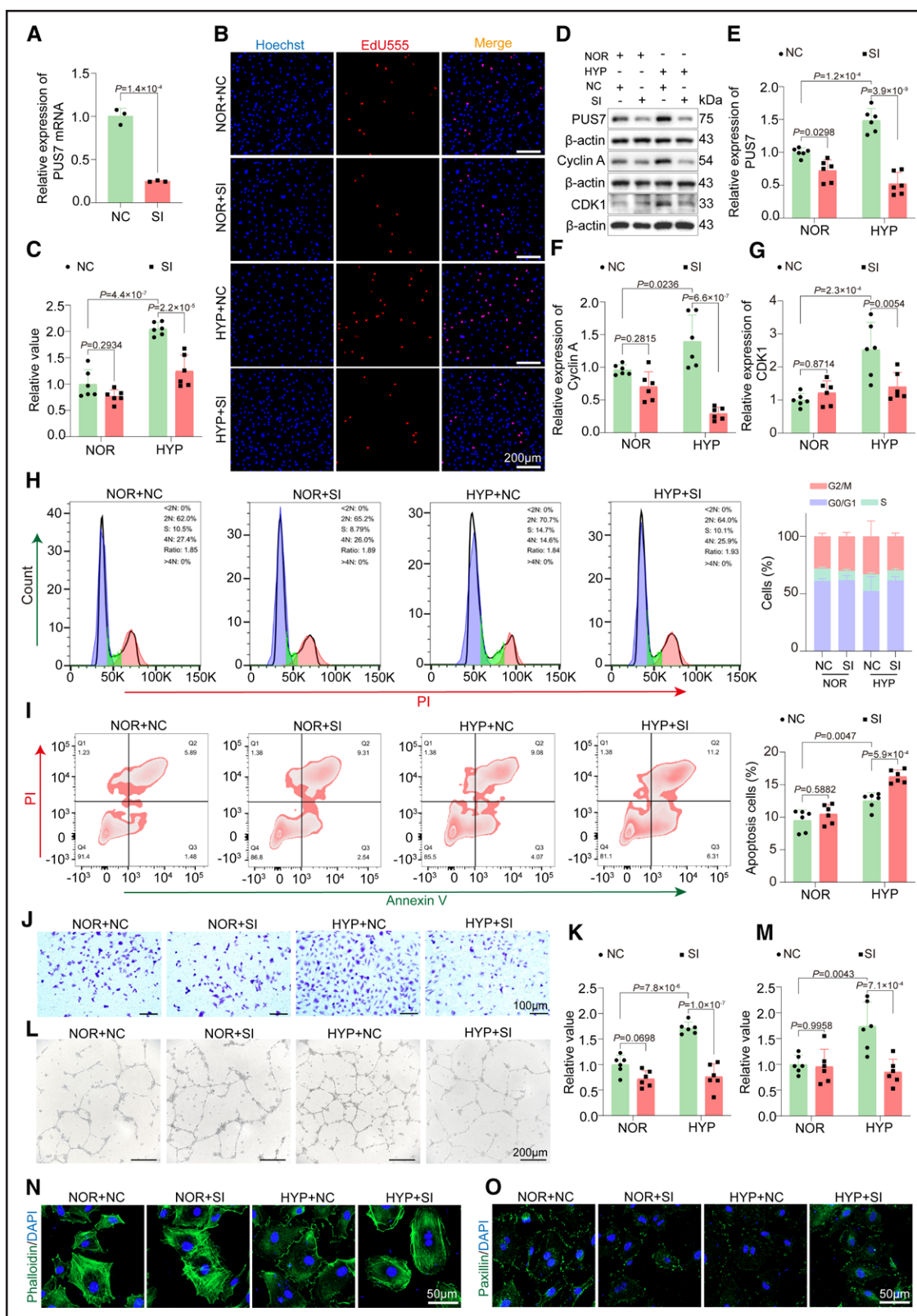


Figure 5. PUS7 deficiency alleviated hypoxia-induced dysfunction of PAECs.

A, Quantitative real-time polymerase chain reaction analysis showing the efficiency of siRNA (SI) of pseudouridine synthase 7 (PUS 7) in pulmonary artery endothelial cells (PAECs; $n=3$ biological replicates). **B** and **C**, Representative EdU staining image and group data showing that inhibition of PUS7 decreased the proliferation of PAECs caused by hypoxia (HYP; $n=6$ biological replicates). **D** through **G**, Representative Western blots and group data showing the expression of PUS7, cyclin A, and CDK1 in PAECs after silencing of PUS7 under hypoxia conditions ($n=6$ biological replicates). **H** and **I**, Flow cytometry analysis showing the effect of inhibition of PUS7 on cell cycle (**H**) and (Continued)

Figure 5 Continued. apoptosis (**I**) in hypoxia-treated PAECs (n=6 biological replicates). **J** and **K**, Transwell analysis showing the migratory capacity of PAECs after PUS7 silencing under hypoxia conditions (n=6 biological replicates). **L** and **M**, Tube formation analysis showing the angiogenic capacity of PAECs after PUS7 knockdown under hypoxia conditions (n=6 biological replicates). **N** and **O**, Confocal microscopy showing the effect of PUS7 knockdown on cytoskeletal architecture of PAECs. Cells were stained with phalloidin (**N**) or paxillin (**O**). Statistical analysis was performed with 2-way ANOVA followed by the Tukey honestly significant difference test for multiple comparisons (**C** through **M**) or unpaired 2-tailed Student *t* test (**A**). All values are presented as mean±SD. NC indicates negative control; and NOR, normoxia.

We further analyzed PUS7-mediated Ψ modification potential sites in TGFBI through deletion ratio analysis from BID-seq, revealing top-scoring sites (positions 628, 688, 844, and 847). We therefore constructed mutant and WT plasmids of TGFBI (Figure 6M). CCK8 analysis showed that TGFBI overexpression for 48 hours rescued PUS7 knockdown-induced reductions in cell viability, but this effect was abrogated by TGFBI mRNA harboring mutations at position 688 (Figure 6N). Actinomycin D chase assays revealed that the U688C mutation significantly shortened TGFBI mRNA half-life compared with WT. Pseudouridylation-deficient mutants exhibited accelerated degradation kinetics, with half-life reductions of $\approx 40\%$ to 50% relative to WT controls (Figure S7E). These findings establish that pseudouridylation at position 688 is necessary for maintaining TGFBI mRNA stability, providing direct functional validation that this specific epitranscriptomic modification confers posttranscriptional regulatory control.

We further studied whether PUS7 modulates TGFBI splicing by Ψ modification of its pre-mRNA. As shown in the exon use schematic in Figure 6O, SI-negative control (red) exhibited significantly higher expression than SI-PUS7 (blue) across exons E010, E011, E023, E027, and E031. These results suggest PUS7 knockdown suppresses these exons, likely by reducing their inclusion in transcripts through altered splicing. In addition, rmats2sashimplot visualization showed that SI-PUS7 significantly altered TGFBI alternative 3' splice site and retained intron events (Figure 6P and 6Q). These results indicate PUS7 enhances TGFBI pre-mRNA splicing efficiency and accelerates its maturation into functional mRNA.

HIF-2 α /PUS7/TGFBI/AKT Signaling Axis Progressively Aggravates PH

EdU and flow cytometry analysis showed that silencing of TGFBI abolished the accelerated cell proliferation and cell cycle progression and increased apoptosis of PAECs induced by PUS7 overexpression (Figure 7A–7C). Transwell (Figure 7D) and tube formation (Figure 7E) assays confirmed that silencing of TGFBI reversed PUS7 overexpression-induced cell migration and angiogenesis. Western blotting analysis revealed that PUS7 overexpression significantly upregulated the expression of phosphorylated AKT and HIF-2 α , but this was blocked by silencing of TGFBI (Figure 7F and 7G). In addition, PUS7 silencing in PAAFs also blunted hypoxia-induced

upregulation of TGFBI and AKT phosphorylation (Figure S7F). Of note, HIF-2 α knockdown abrogated hypoxia-induced PUS7 upregulation, suggesting that HIF-2 α and PUS7 may regulate each other (Figure 7H and 7I). In silico analysis using hTFtarget (<https://guolab.wchscu.cn/hTFtarget/#/prediction>) revealed that HIF-2 α binds to the PUS7 promoter (Figure 7J). Dual-luciferase reporter assays showed that cotransfection of HIF-2 α and WT PUS7 promoter plasmids significantly increased luciferase activity, whereas mutations at predicted binding positions abrogated this effect (Figure 7K). Chromatin immunoprecipitation assays with primers targeting 2 predicted loci showed that only the position 1 (P1) region was amplified from HIF-2 α chromatin immunoprecipitants, confirming specific HIF-2 α binding to the PUS7 promoter P1 region (Figure 7L). Collectively, these results demonstrate that PUS7 mediates PAEC dysfunction by upregulation of TGFBI and its downstream PI3K-AKT signaling to drive progressive PH aggravation.

Of note, experiments using recombinant protein treatment and conditioned medium assays demonstrate that TGFBI serves as the predominant paracrine mediator responsible for endothelial PUS7-driven smooth muscle remodeling in small pulmonary arteries (Figure S8). Collectively, these findings support the following model: Endothelial PUS7 upregulation promotes TGFBI secretion from PAECs, which subsequently activates AKT signaling in adjacent PSMCs to drive proliferation and vascular remodeling. This paracrine PUS7-TGFBI axis thus represents a critical endothelial-smooth muscle communication pathway in PH pathogenesis.

Inhibition of PUS7 With NSC107512 Ameliorates PH Development in Animal Models

NSC107512 is a small-molecule inhibitor of PUS7. PyMOL and Discovery Studio analyses identified the binding domain and interacting amino acid residues in PUS7 (Figure 8A). CCK8 assays demonstrated that NSC107512 acted in a concentration-dependent manner, with 40 nmol/L eliciting a marked antiproliferative response (Figure S9A). Western blotting analysis showed that NSC107512 attenuated hypoxia-induced TGFBI upregulation (Figure 8B). EdU and flow cytometry assays revealed that NSC107512 abolished hypoxia-accelerated proliferation and cell cycle progression in PAECs (Figure 8C and 8D).

Hemodynamic, echocardiographic, and ELISA analyses demonstrated that NSC107512 exerted

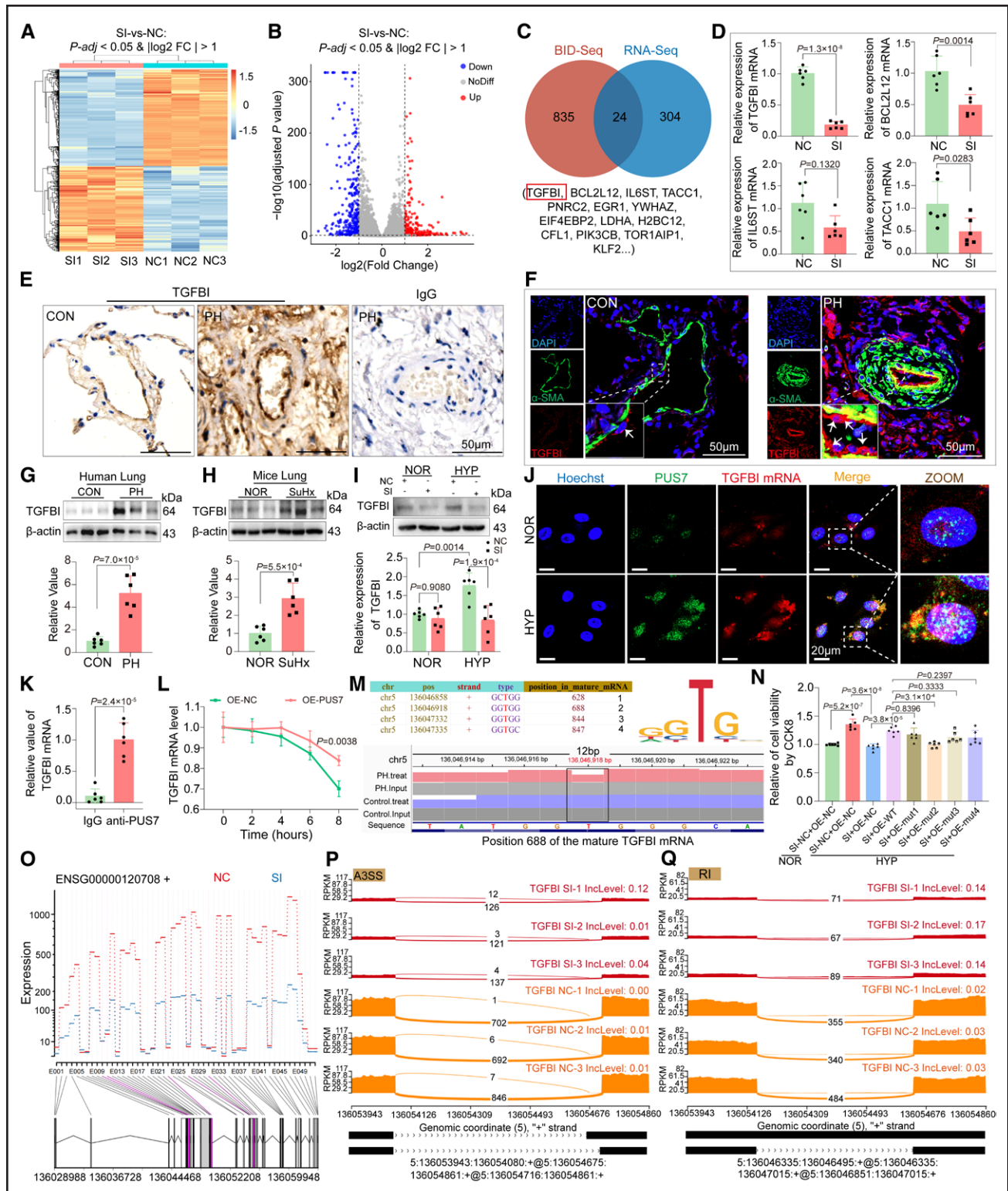


Figure 6. PUS7 regulated PAEC functions via Ψ modification of the TGFBI mRNA.

A and **B**, Heat maps (**A**) and volcano plots (**B**) depicting pseudouridine synthase 7 (PUS 7) loss resulted in differential gene expression in pulmonary artery endothelial cells (PAECs). **C**, Venn diagrams showing the top overlapping genes between bisulfite-induced deletion sequencing (BID-seq) and RNA high-throughput transcriptome sequencing (RNA-seq). **D**, Quantitative real-time polymerase chain reaction analysis showing the candidate gene expression in PAECs after PUS7 knockdown (n=6 biological replicates). **E** and **F**, Immunohistochemistry (**E**) and immunohistochemical staining (**F**) showing that compared with healthy donors, transforming growth factor- β induced protein (TGFBI) was highly expressed in the pulmonary artery intima of patients with pulmonary hypertension (PH). **G** and **H**, Western blotting analysis showing the expression of TGFBI in the lungs of patients with PH (**G**; n=6 patients) and hypoxia (HYP) combined with SU5416 (SuHx) PH (**Continued**)

Figure 6 Continued. mouse models (**H**; $n=6$ male mice). **I**, Representative Western blots and group results showing the expression of TGFBI in PAECs after knockdown of PUS7 under hypoxia conditions ($n=6$ biological replicates). **J**, Representative confocal images of fluorescence in situ hybridization staining for PUS7 (green) and fluorescent probe for TGFBI mRNA (red) in hypoxic PAECs. Nuclei were stained with Hoechst (blue). **K**, RNA immunoprecipitation assay showing that PUS7 can directly bind to TGFBI mRNA ($n=6$ biological replicates). **L**, Actinomycin D assay showing that overexpression (OE) of PUS7 stabilizes the level of TGFBI mRNA ($n=6$ biological replicates). **M**, Plot depicts pseudouridine (Ψ) modification sites in TGFBI mRNA as detected by BID-seq. **N**, CCK8 analysis showing the cell viability of PAECs after PUS7 siRNA cotransfected with expression plasmids harboring either the Ψ modification site mutant or wild-type TGFBI mRNA under hypoxia conditions ($n=6$ biological replicates). **O**, Differences in exon usage in RNA high-throughput transcriptome sequencing (RNA-seq) data analyzed by the DEXSeq package; x axis shows exon positions; y axis show the exon expression of 2 groups. **Top Right**, Color coding for groups. Fuchsia on x axis indicates significant exon use differences between groups. **P** and **Q**, Sashimi plots illustrate alternative 3' splice site (A3SS; **P**) and retained intron (RI; **Q**) events of TGFBI in control (CON; negative control [NC]; orange) and PUS7 silencing (siRNA of PUS7 [SI]; red) samples. Statistical analysis was performed with 1-way (**N**) or 2-way ANOVA followed by the Tukey honestly significant difference test for multiple comparisons (**I**), unpaired 2 tailed Student *t* test (**D**; TGFBI, BCL2L12, TACC1; **G**, **H**, **K**, **L**), or 2-tailed Mann-Whitney test (**D**; IL6ST). All values are presented as mean $\pm$ SD. NOR indicates normoxia.

dose-dependent therapeutic effects and minimal systemic toxicity in vivo (Figure S9B–S9H). We further evaluate the therapeutic potential of PUS7 inhibition on PH in a SuHx-induced PH rat model (Figure 8E). NSC107512 did not significantly affect rat body weight (Figure 8F and Figure S10A). Echocardiography (Figure 8G–8I) and hemodynamic measurements (Figure 8J) showed that NSC107512 attenuated SuHx-induced reductions in pulmonary artery velocity-time integral, pulmonary artery acceleration time/ejection time ratio, and tricuspid annular plane systolic excursion, as well as increased RV systolic pressure, mean pulmonary artery pressure, and RV/left ventricular plus septal weight (Figure S10B and S10C). Immunostaining (Figure 8K) and hematoxylin-eosin staining (Figure 8L–8N) analysis revealed that NSC107512 reduced SuHx-induced vascular remodeling (thickened vascular walls, increased muscularization, expanded α -smooth muscle actin–positive areas, neointimal formation) and RV hypertrophy. Western blotting analysis confirmed that in vivo NSC107512 administration reversed SuHx-induced TGFBI and phosphorylated AKT upregulation (Figure 8O and Figure S10D). Similar results were also found in bleomycin-induced PH rats (Figure S10E–S10R). These results suggest that by targeting PUS7, NSC107512 is a new potential therapeutic agent to treat PH.

DISCUSSION

PH remains to be a recalcitrant disorder characterized by unrelenting vascular remodeling, with limited therapeutic options targeting the underlying pathogenic mechanisms. In this study, we identified PUS7-mediated pseudouridylation as a novel epigenetic driver of PH, unraveling an HIF-2 α /PUS7/TGFBI/PI3K-AKT signaling axis that propagates PAEC dysfunction and vascular remodeling. Our findings not only expand the landscape of RNA modifications in PH pathogenesis but also highlight PUS7 as a potential therapeutic target, offering new insights into disease mechanisms and intervention strategies.

Although a case report described severe PH in a patient with congenital sideroblastic anemia secondary

to a PUS1 gene mutation,²⁵ the link between Ψ and PH remains poorly defined. To address this knowledge gap, we used BID-seq, a cutting-edge technique that enables single-base resolution mapping of Ψ modifications by leveraging the unique chemical reactivity of Ψ with bisulfite, overcoming the limitations of traditional methods, which lack precision in quantifying individual modification sites.^{26,27} We mapped the first single-base resolution Ψ landscape in PH lung tissues, revealing global dysregulation of Ψ modifications with enrichment in pathways central to vascular remodeling, including smooth muscle contraction, extracellular matrix organization, and PI3K-AKT signaling. This widespread epigenetic reprogramming suggests that Ψ modifications are not passive bystanders but active regulators of gene expression in PH.

Among several well-characterized PUSs, we found that only PUS7 exhibited upregulation in tissues and PAECs from patients with PH exposed to hypoxia, indicating it as a key mediator of pathological Ψ modification. In vitro results demonstrated that increased PUS7 expression drove PAEC dysfunction, whereas PUS7 knockdown elicited the opposite effect. Consistently, PUS7 overexpression accelerated the pathological progression of PH, whereas PUS7 knockdown attenuated hypoxia-, SuHx-, or bleomycin-induced PH-related pathological changes. Although pulmonary arterial hypertension and group 3 PH have distinct initiating causes, they converge on common pathological alterations: Endothelial and smooth muscle cell dysfunction drives intimal and medial thickening and muscularization, ultimately elevating pulmonary vascular resistance.³ Our data demonstrate that PUS7 is significantly upregulated in both subtypes and mediates PAEC dysfunction through the TGFBI/phosphorylated AKT signaling axis. These findings suggest that PUS7-mediated pseudouridylation may represent a conserved epigenetic checkpoint in vascular remodeling, a unifying mechanism across PH subtypes that could inform broadly applicable therapeutic strategies.

TGFBI, a matricellular protein involved in cell adhesion and extracellular matrix interactions, contributes to the pathophysiology of pulmonary arterial hypertension

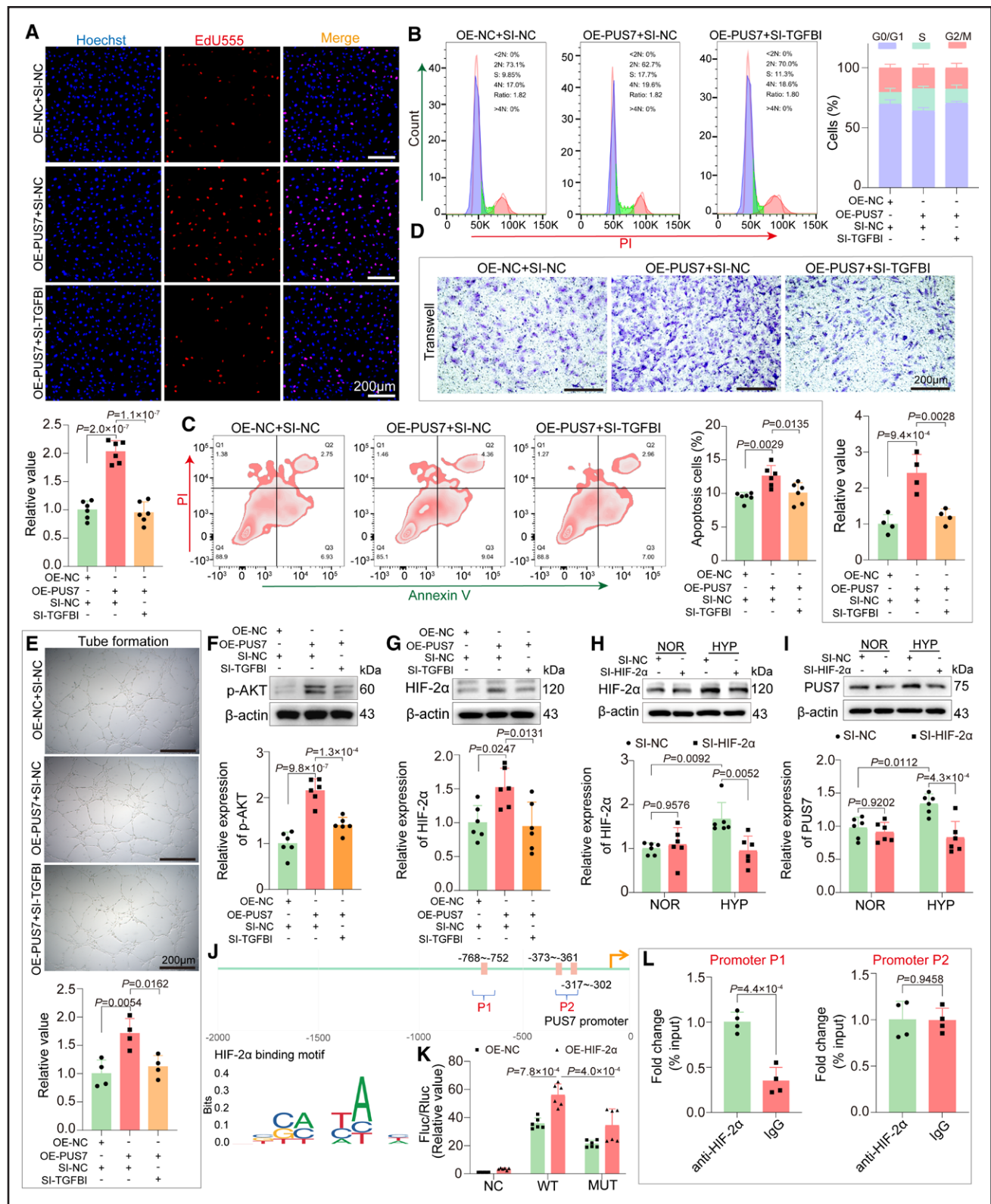


Figure 7. HIF-2α/PUS7/TGFβ1/PI3K-AKT signaling axis progressively aggravated pulmonary hypertension.

A through **E**, Silencing of SI-TGFβ1 (siRNA-transforming growth factor β-induced protein) abolished the effects of pseudouridine synthase 7 (PUS7) overexpression (OE) in pulmonary artery endothelial cells (PAECs). **A**, EdU assay: cell proliferation (n=6 biological replicates); **B** and **C**, Flow cytometry analysis: cell cycle (**B**) and apoptosis (**C**; n=6 biological replicates); **D**, Transwell assay: migratory capacities (n=4 biological replicates); **E**, Tube formation analysis: angiogenic capacity (n=4 biological replicates). **F** through **I**, Representative Western blots and quantification showing the effect of silencing of SI-TGFβ1 on the expression of phosphorylated AKT (p-AKT; **F**) and hypoxia-induced factor-2α (HIF-2α; **G**; n=6 biological replicates) and the effect of SI-HIF-2α (**H**) on PUS7 (**I**) in PAECs under normoxia (NOR) or (Continued)

Figure 7 Continued. hypoxia (HYP) conditions (n=6 biological replicates). **J**, hTFtarget analysis showing the binding motif of HIF-2 α to the PUS7 promoter region. **K**, Dual luciferase vector and mutant plasmids were designed according to binding sites with the HIF-2 α and PUS7 promoter. Dual luciferase assays were used to validate the interactions between HIF-2 α and PUS7 promoter in 293T cells (n=6 biological replicates). **L**, Chromatin immunoprecipitation–quantitative real-time polymerase chain reaction analysis showing that the PUS7 promoter region was detected in fractions immunoprecipitated with anti-HIF-2 α (n=4 biological replicates). Statistical analysis was performed with 1-way ANOVA analysis followed by the Tukey honestly significant difference multiple comparison test (**A** through **G**) or 2-way ANOVA (**H**, **I**, **K**) followed by the Tukey honestly significant difference test for multiple comparisons or unpaired 2-tailed Student *t* test (**L**). All values are presented as mean $\pm$ SD. NC indicates negative control.

and group 4 PH (chronic thromboembolic PH) by mediating vascular remodeling through endothelial-to-mesenchymal transition and promoting impaired resolution and fibrosis of venous thrombi.^{28,29} Our studies demonstrated that PUS7 bound directly to TGFBI mRNA, catalyzing site-specific Ψ modifications (position 688) to enhance its stability. RIP-sequencing analysis revealed that PUS7 binds to multiple sites across the TGFBI transcript, including both intronic and exonic regions. Notably, robust binding signal was observed at exon 5, which precisely colocalizes with the pseudouridylation site identified in our BID-seq data. This concordance between independent methodologies—RIP sequencing detecting PUS7 protein-RNA binding and BID-seq detecting Ψ modification sites—provides robust evidence for direct, site-specific PUS7-mediated pseudouridylation of TGFBI mRNA.

The presence of PUS7 binding in exonic regions suggests potential cotranscriptional or cotranslational regulation, whereas intronic binding at the BID-seq-identified site supports a direct role in modulating mRNA splicing and stability. Consistently, our RNA high-throughput transcriptome sequencing data showed that PUS7 knock-down disrupted TGFBI pre-mRNA alternative 3' splice site and retained intron. These findings substantially strengthen our mechanistic conclusion that TGFBI is a bona fide, selective target of PUS7 in PAECs and indicate that PUS7 enhances TGFBI mRNA maturation and stability by regulating splicing efficiency. Collectively, this posttranscriptional regulation is functionally significant: TGFBI upregulation activates PI3K-AKT signaling, a pathway known to drive PAEC proliferation, migration, and angiogenesis, hallmarks of PH-associated endothelial dysfunction.³⁰

Furthermore, we uncovered a positive feedback loop amplifying this axis: Activation of PI3K-AKT-mTORC2 stabilizes HIF-2 α ,^{31,32} a hypoxia-inducible transcription factor that directly binds to the PUS7 promoter to enhance its expression. This loop creates a self-reinforcing cycle wherein hypoxia-induced HIF-2 α upregulates PUS7, which in turn stabilizes TGFBI to activate PI3K-AKT, further promoting HIF-2 α accumulation. This mechanism provides an explanation for the progressive nature of PH; pathological stimuli initiate a cascade that perpetuates vascular remodeling even after the initial trigger is removed. The enrichment of PUS7 in PAECs aligns with its role in endothelial dysfunction, rendering it a potential biomarker for vascular lesions.

NSC107512 is a small molecule that selectively binds the catalytic domain of PUS7 and inhibits its activity.²⁴ We validated in the present study that NSC107512 attenuated PH progression in SuHx and bleomycin models, reducing RV systolic pressure, vascular wall thickening, and RV hypertrophy. A previous study demonstrated that NSC107512 exhibits minimal off-target potential and demonstrated safety in animal PH models, supporting its translational potential.²⁴ Unlike current PH therapies that primarily target vasodilation, NSC107512 addresses the root cause of disease by halting vascular remodeling, representing a paradigm shift in PH treatment.

RNA modifications have emerged as critical modulators of cardiovascular diseases, including PH.^{33,34} We previously identified ADAR1-mediated A-to-I RNA editing and reactive oxygen species-driven 8-oxoguanine modifications as key regulators of pathological vascular remodeling in PH.^{35,36} Although N⁶-methyladenosine promotes PASMC hyperproliferation and endothelial-to-mesenchymal transition,^{37,38} Ψ offers distinct advantages: Its stable C-C glycosidic bond enhances RNA hydrolysis resistance and stability^{16,39}; its non-Watson-Crick edge provides an additional hydrogen bond donor for stronger base pairing; and its irreversible nature avoids regulatory complexity. These properties position Ψ as a promising candidate for RNA regulation, cardiovascular gene therapy, and drug development.

Several limitations exist in this study. First, the broader expression pattern of PUS7 explains detectable signals in nonmuscular cells, suggesting potential roles in lung parenchymal remodeling requiring further investigation. Second, abundant patient-isolated PH-PAECs are difficult to obtain, prompting us to use commercially sourced human PAECs for functional and mechanistic characterization. Nevertheless, the acquired results remain informative to decipher the biological function and underlying mechanism of PUS7 in PH development. Third, although NSC107512 shows target specificity, complete off-target elimination remains challenging. Future chemoproteomics validation will further confirm target engagement.

CONCLUSIONS

This study identifies PUS7-mediated pseudouridylation as a novel epigenetic driver of PH through an HIF-2 α /PUS7/TGFBI/PI3K-AKT positive feedback loop. PUS7 inhibition emerges as a potential therapeutic strategy

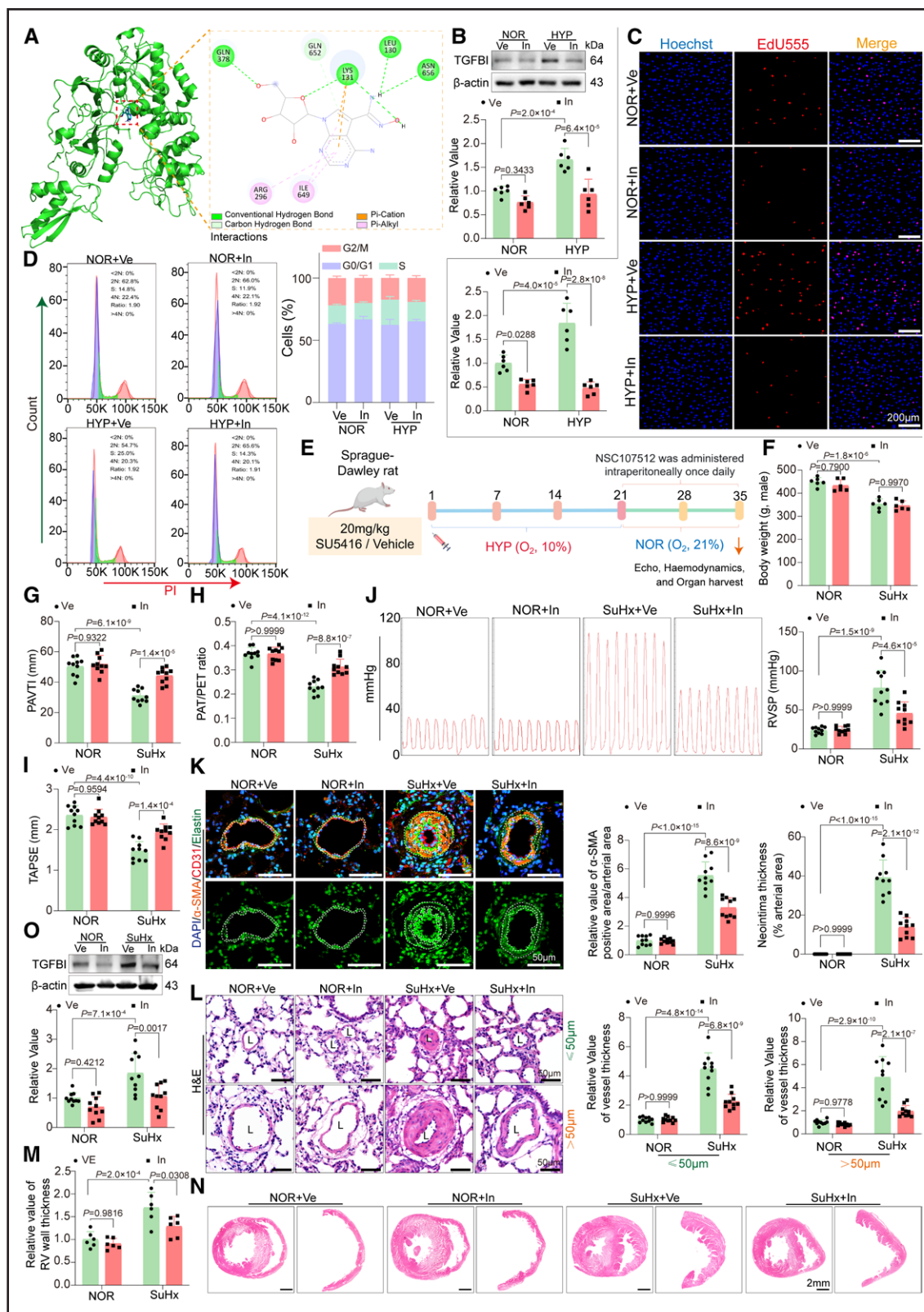


Figure 8. Effect of NSC107512 on PH in animal models.

A, PyMOL and Discovery Studio were utilized to map the binding domain and specific amino acids involved in the interaction between NSC107512 and pseudouridine synthase 7 (PUS7) protein. **B**, Representative Western blots and group data showing that inhibition of PUS7 alleviated the increased expression of transforming growth factor β -induced protein (TGFB1) induced by hypoxia (HYP) in pulmonary artery endothelial cells (PAECs; n=6 biological replicates). **C**, EdU analysis showing that inhibition of PUS7 by NSC107512 decreased the proliferation of PAECs caused by hypoxia (n=6 biological replicates). **D**, Flow cytometry analysis showing the effect of NSC107512 on cell (Continued)

Figure 8 Continued. cycle in hypoxia-treated PAECs (n=6 biological replicates). **E**, Schematic protocol for the animal treatments in the hypoxia combined with SU5416 (SuHx)-induced pulmonary hypertension (PH) rat model. **F**, Body weight of SuHx-induced PH rat treated with NSC107512 (n=6 male rats). **G** through **I**, Group analysis showing that inhibition of PUS7 by NSC107512 significantly attenuated the decreased pulmonary artery velocity-time integral (PAVTI; **G**), pulmonary artery acceleration time/ejection time ratio (PAT/PET; **H**), and tricuspid annulus plain systolic excursion (TAPSE; **I**) in rats treated with SuHx (n=10 rats, 6 males and 4 females). **J**, Representative hemodynamic tracings and group analysis showing the therapeutic effects of PUS7 inhibition by NSC107512 on right ventricular systolic pressure (RVSP) of the SuHx rat model (n=10 rats, 6 males and 4 females). **K** through **N**, Immunofluorescence and hematoxylin-eosin (H&E) staining showing that inhibition of PUS7 by NSC107512 significantly ameliorated the SuHx-induced increases in vascular muscularization and neointimal formation (**K**; n=10 rats, 6 males and 4 females), medial wall thickness (**L**; n=10 rats, 6 males and 4 females), and right ventricular wall thickness (**M** and **N**; n=6 male rats) of the SuHx rat model. **O**, Representative Western blots and group data showing that inhibition of PUS7 alleviates the increased expression of TGF β 1 induced by SuHx in lung tissues of the PH rat model (n=10 rats, 6 males and 4 females). Statistical analysis was performed with 2-way ANOVA followed by the Tukey honestly significant difference test for multiple comparisons (**B** through **O**); All values are presented as mean $\pm$ SD. In indicates inhibitor of PUS7 (NSC107512); L, lumen; NOR, normoxia; and Ve, vehicle.

to reduce vascular remodeling, offering a foundation of epitranscriptomic-based PH treatments.

ARTICLE INFORMATION

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Affiliations

Department of Pharmacology, Joint Laboratory of Guangdong–Hong Kong Universities for Vascular Homeostasis and Diseases (J.Z., Y.L., M.H., Z.L., H.W., F.L., X.C., Z.J., H.C., Z.C., F.K., J.F., J.-S.B.), and Department of Human Cell Biology and Genetics (X.-W.N.), SUSTech Homeostatic Medicine Institute, School of Medicine, Southern University of Science and Technology, Shenzhen, China. Shenzhen Institute of Respiratory Disease, Shenzhen Key Laboratory of Respiratory Disease, Shenzhen People's Hospital (The Second Clinical Medical College, Jinan University; The First Affiliated Hospital, Southern University of Science and Technology), China (Y.L.). Post-Doctoral Scientific Research Station of Basic Medicine, Jinan University, Guangzhou, China (Y.L.). Lung Transplant Group, Wuxi People's Hospital Affiliated to Nanjing Medical University, Wuxi, China (J.T., Y.C.).

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Disclosures

None.

Supplemental Material

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Database links

Single-cell transcriptome sequencing: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE335228>
Bulk RNA sequencing: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE334780>
RNA BID-seq: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE335339>
RIP sequencing: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE334990>

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